

The circulated calibration note

What reproduces, what does not, and what M5 can currently support

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Audit result

The circulated parameter and moment tables reproduce the canonical M5 solution exactly.

14 / 14

parameter estimates

15 / 15

model moments

9 / 14

effective rank at 1%

The numerical result is reproducible. The structural interpretation is substantially less secure.

Source: M5 strict repeated winner and local-identification audit.

Model architecture

M5 links family needs to housing access and lifecycle balance sheets.

Family formation

- Completed family size chosen once
- One shared child-maturity clock
- No sequential birth hazards

Housing and tenure

- Rent continuously up to 6 rooms
- Own one of $\{2, 4, 6, 8, 10\}$ rooms
- Purchase requires $(1 - \phi)PH = 0.2PH$

Saving and bequests

- Liquid wealth may be negative
- Terminal estate is $b + PH$
- No amortizing mortgage contract

The model is appropriate for completed-fertility and housing-access mechanisms, not for birth timing or mortgage-contract claims.

Source: Active M5 implementation and implementation ledger.

Calibration design

Estimated jointly

- 14 preference, needs, tenure, supply, and bequest parameters
- 15 fertility, housing, ownership, wealth, and estate moments
- Weighted loss: 9.044

Maintained or external

- Four-year timing, $\sigma = 2$, and housing-finance rates
- Income persistence and low innovation risk
- $\theta_n = 0$ and one national housing market

The loss mixes inverse-variance and judgmental weights. It ranks candidates under this target system; it is not a chi-squared statistic.

Source: M5 parameter table, target-fit table, and objective definition.

Local identification

9 / 14

effective rank at 10^{-2}

12 / 14

effective rank at 10^{-3}

2,926

condition number

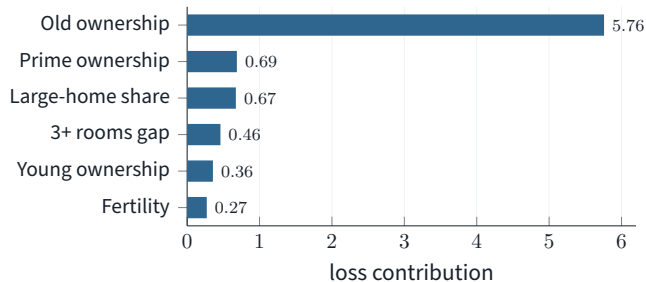
The weakest direction jointly loads on

$$(\bar{h}_0, \theta_1, \theta_0, \kappa_T).$$

The moment-to-parameter discussion in the note is economic intuition, not evidence of one-to-one identification.

Source: SMM-weighted Jacobian audit.

Loss decomposition



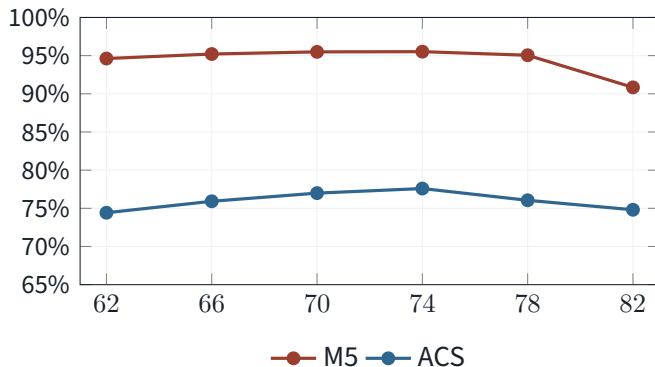
63.7%

of total loss comes from
ownership at ages 65–75.

The estate median is
close to its target;
late-life tenure is not.

Source: M5 target-fit table.

Late-life ownership



95.4%

M5, ages 65–75

76.4%

ACS target

Liquid wealth turns negative while ownership remains above 90%.

Source: M5 lifecycle diagnostics and ACS ownership profile.

Fit by empirical margin

Close to target

- Completed childlessness
- First-child housing response
- Median estate and older nonhousing wealth
- Aggregate occupied rooms

Binding misses

- Old ownership: +15.1 bootstrap SE
- Large-home share: +9.4 SE
- Family ownership gap: +8.0 SE
- Prime ownership: +4.1 SE
- Young ownership: -3.7 SE

The bootstrap ratios are descriptive, not a joint specification test, because the active objective does not use the full covariance matrix.

Source: M5 fit and July 6 bootstrap standard-error audit.

Target provenance

Two headline targets are not presently reproducible from a repository builder:

completed fertility = 1.918, childlessness = 0.188.

The note says CPS, but the year, age window, children-ever-born variable, and weights are not pinned down.

If both targets were removed without replacement, the calibration would have only 13 verified moments for 14 free parameters.

They must be sourced, replaced by equally informative moments, or paired with an external restriction.

Source: July 6 target-reproduction audit.

Evidence after circulation

Income frontier

- The M5 bundle is infeasible already at $\sigma_z = 0.09$
- Lowering $\bar{c}_0$, not a small transfer floor, moves the frontier

Transfer-floor probe

- Honest risk raises the estate tail
- Young wealth rises to 1.33, far above the 0.18 target

E1 experiment

- Repairs late-life ownership and the estate tail
- Worsens fertility and first-child housing fit
- Loss rises from 9.04 to 12.61

These exercises clarify the trade-offs; none is a replacement baseline.

Source: July 18 frontier and floor probes; July 19 E1 collection report.

Defensible interpretation

M5 is a reproducible, provisional calibration for studying how family housing needs interact with tenure and down-payment access.

It does **not** currently support:

- precise estimates of all 14 structural parameters;
- claims about sequential birth timing or mortgage contracts;
- a well-identified late-life portfolio or housing-exit mechanism;
- formal specification testing based on the scalar loss.

The next baseline should be judged by whether it preserves the housing-fertility mechanism while repairing provenance, identification, and late-life behavior.

Questions for the next baseline

1. Is one-shot completed fertility acceptable if timing claims are removed?
2. Should the next baseline prioritize income insurance or late-life housing exit?
3. Should weak bequest and tenure parameters be externally fixed or newly identified?
4. Which paper claims actually require a credible late-life portfolio block?

M5 parameter estimates

Parameter	Estimate	Bounds	Status	Parameter	Estimate	Bounds	Status
β_{ann}	.9912	[.800,.9995]	est.	ψ_{child}	.1979	[-3,3]	est.
α_c	.5912	[.02,.98]	est.	κ_F	2.0434	[.02,50]	est.
$\bar{c}_0$	1.2596	[0,2]	est.	χ_O	1.1133	[.1,5]	est.
$\bar{c}_n$	.3929	[0,3]	est.	$\bar{H}$	8.6456	[.2,80]	est.
$\bar{h}_0$	.3915	[.05,5.8]	est.	θ_0	.3118	[0,8]	est.
$\bar{h}_{jump}$	1.6021	[0,8]	est.	θ_1	.3973	[.02,16]	est.
$\bar{h}_n$	.1644	[0,5]	est.	κ_T	.0100	[0,.12]	est.
				θ_n	0	external	fixed

No estimate is mechanically near its search bound. That does not imply precise identification.

Source: M5 parameter collector.

Target fit: fertility, housing, and young wealth

Moment	Data	Model	Gap	Weight	Loss
Completed-fertility-equivalent*	1.918	2.034	.116	20.0	.271
Completed childlessness*	.188	.189	.001	20.0	.000
Ownership, ages 30–55	.575	.658	.083	100.0	.687
New-parent ownership gap	.168	.219	.052	45.0	.121
First-child rooms response	.664	.681	.016	14.0	.004
Childless-renter mean rooms	3.805	3.963	.158	6.0	.150
Childless-owner share, 6+ rooms	.596	.760	.164	25.0	.673
Young liquid wealth / income	.179	.328	.149	12.0	.265

* Source-unconfirmed in the repository reproduction audit.

Target fit: tenure, estates, and aggregates

Moment	Data	Model	Gap	Weight	Loss
Owner-renter rooms gap	2.419	2.323	-.096	12.0	.111
Ownership, ages 25-34	.341	.274	-.067	80.0	.356
Rooms gap, 3+ versus 1-2 children	.368	.607	.240	8.0	.460
Median estate / income	6.501	6.431	-.070	18.6	.092
Nonhousing wealth $\geq$ income	.608	.610	.002	9435.2	.027
Ownership, ages 65-75	.764	.954	.190	160.0	5.759
Mean occupied rooms	5.780	5.673	-.107	6.0	.069
Total					9.044

Source qualifications

- The PDF circulated on July 17 reports a two-plus-versus-one-child estate gap of 0.101. The later source adds that it is statistically insignificant (bootstrap SE 0.563).
- The later source also notes that PSID nonhousing wealth includes business wealth, retirement accounts, and other assets absent from M5.
- External values were checked against the 2023 SSA life table, Greaney, Parkhomenko, and Van Nieuwerburgh (2025), and Saiz (2010).

The circulated PDF, not the subsequently edited TeX file, is the record of what recipients saw.